Hints: Polynomials – 3. Symmetric functions, Newton's identities, resultant and discriminant

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. The first sum is antisymmetric: the terms for (i, j) and (j, i) cancel.

Hint 2. Write $P = (x - x_i)R_i$. Then $P'(x_i) = R_i(x_i)$ and $P''(x_i) = 2R_i'(x_i)$, so $\frac{P''(x_i)}{P'(x_i)} = 2\sum_{j\neq i} \frac{1}{x_i - x_j}$; sum over i and use the first part.

Hint 3. From $\frac{P'(x)}{P(x)} = \sum_i \frac{1}{x - x_i}$: $\sum_i \frac{1}{x_i - a} = -\frac{P'(a)}{P(a)}$.

Problem 2

Hint 1. $F(z) = \prod(1 - z^{k_j})$ is divisible by $(1 - z)^n$, so $F^{(s)}(1) = 0$ for $s < n$, i.e. $\sum_m c_m\, m^s = 0$ for $s = 0, 1, \ldots, n - 1$, where $F = \sum c_m z^m$ has integer coefficients.

Hint 2. Since $F(1) = 0$, the positive coefficients and the negative coefficients have the same total N, and the sum of absolute values is $2N$. Form the multiset A containing each m exactly c_m times when $c_m > 0$, and B containing m exactly $|c_m|$ times when $c_m < 0$; both have N elements and disjoint supports.

Hint 3. Hint 1 says A and B have equal power sums p_s for $s = 1, \ldots, n - 1$. If $N \leqslant n - 1$, Newton's identities give equal $e_1, \ldots, e_N$, so $A = B$ as multisets (same polynomial $\prod(t - a)$), which is impossible. Hence $N \geqslant n$.

Problem 3

Hint 1. Consider the rational function $R(x) = \dfrac{p(x + 1)}{x\, p(x)}$. Its poles are $0, \alpha_1, \ldots, \alpha_n$, all simple. What are the coefficients of its partial fraction expansion?

Hint 2. The coefficient at $x = 0$ is $\frac{p(1)}{p(0)}$ and at $x = \alpha_i$ it is $\frac{p(\alpha_i + 1)}{\alpha_i p'(\alpha_i)}$.

Hint 3. For a rational function $\sum_k \frac{c_k}{x - \beta_k}$ + (polynomial) the sum $\sum c_k$ is the coefficient of $1/x$ at infinity, i.e. $\lim_{x\to\infty} x \cdot$ (proper part). Here $R(x)$ is proper and $xR(x) \to 1$.

Problem 4

Hint 1. Same idea as Problem 3: find a rational function whose partial fraction coefficients are exactly the terms of the sum. Try $R(z) = \dfrac{1}{A(z)A(\alpha z)}$.

Hint 2. The poles z_k and z_k/α are simple and pairwise distinct (no ratio of zeros equals α). The coefficient at z_k is $\frac{1}{A'(z_k)A(\alpha z_k)}$; at z_k/α it is $\frac{1}{\alpha A'(z_k)A(z_k/\alpha)}$, since $\frac{d}{dz}A(\alpha z) = \alpha A'(\alpha z)$.

Hint 3. $R(z) = O(z^{-2n})$ with $n \geqslant 1$, so $zR(z) \to 0$ and the sum of all coefficients is 0.

Problem 5

Hint 1. By Vieta $x_1 + x_2 = p$, $x_1 x_2 = -\frac{1}{2p^2}$. Compute $x_1^2 + x_2^2$ and then $x_1^4 + x_2^4 = (x_1^2 + x_2^2)^2 - 2x_1^2 x_2^2$.

Hint 2. You get $x_1^4 + x_2^4 = p^4 + \frac{1}{2p^4} + 2$; finish with AM–GM: $p^4 + \frac{1}{2p^4} \geqslant \sqrt{2}$.

Problem 6

Hint 1. Work in the quotient $\mathbb{R}[X]/M$. From $\prod_j (1 - X_j t) \equiv 1$ modulo M deduce relations involving the complete homogeneous symmetric polynomials h_m of a subset of the variables.

Hint 2. $\prod_{j<i}(1 - X_j t) = \prod_j (1 - X_j t) \cdot \prod_{j \geqslant i} (1 - X_j t)^{-1}$ shows $h_m(X_i, \dots, X_n) \equiv \pm e_m(X_1, \dots, X_{i-1}) \bmod M$, which is 0 for $m \geqslant i$. In particular $h_i(X_i, \dots, X_n) \in M$, and it is monic of degree i in X_i.

Hint 3. Use these relations to rewrite any monomial, modulo M and keeping the degree, as a combination of monomials $X_1^{a_1} \cdots X_n^{a_n}$ with $a_i \leqslant i-1$ (reduce the exponent of X_i using $h_i(X_i, \dots, X_n)$, which only brings in $X_i, \dots, X_n$). Such monomials have degree $\leqslant \sum (i-1) = \binom{n}{2}$, so any monomial of higher degree lies in M.

Hint 4. For the second part, use the Vandermonde $\Delta = \prod_{i<j}(X_j - X_i)$ and the pairing $\langle f, g \rangle = f(\partial) g \,\big|_0$. Since $e_k(\partial)$ is symmetric and $e_k(\partial)\Delta$ is an antisymmetric polynomial of degree $< \binom{n}{2}$, it is 0; so $\langle f, \Delta \rangle = 0$ for every $f \in M$ of degree $\binom{n}{2}$.

Hint 5. But the monomial $X_2 X_3^2 \cdots X_n^{n-1}$ occurs in Δ with coefficient ± 1, so its pairing with Δ is $\pm \prod_k k! \neq 0$. Hence it is not in M.

Problem 7

Hint 1. The S_k are the coefficients of the monic factor $g(T) = \prod_{i \leqslant p}(T - x_i)$ of f over a splitting field L. How many monic factors of degree p does f have over L?

Hint 2. L/K is Galois (f is separable). Every $\sigma \in \mathrm{Gal}(L/K)$ maps g to another monic degree-p factor $\prod_{i \in I}(T - x_i)$ of f, determined by the p-subset I; there are $\binom{n}{p}$ of these.

Hint 3. The subgroup H fixing the set $\{x_1, \dots, x_p\}$ fixes all S_k, so $K(S_1, \dots, S_p) \subseteq L^H$ and $[L^H : K] = [G : H]$ is the size of the orbit of this set, at most $\binom{n}{p}$.

Problem 8

Hint 1. Use $D = a^4 \prod_{i<j}(x_i - x_j)^2$ and examine each case of the roots.

Hint 2. Three distinct real roots: every factor is a positive square. One real root r and a pair $z, \bar{z}$: $(z - \bar{z})^2 = -4(\operatorname{Im} z)^2 < 0$ while $(r - z)^2 (r - \bar{z})^2 = |r - z|^4 > 0$, so $D < 0$. A multiple root gives $D = 0$.

Hint 3. For $x^3 + px + q$ Cardano's formula involves $\sqrt{\frac{q^2}{4} + \frac{p^3}{27}} = \sqrt{-D/108}$. So exactly when all three roots are real and distinct one must take square roots of a negative number (casus irreducibilis): the real roots cannot be reached without complex numbers.

Problem 9

Hint 1. If α is a common root, the vector $(\alpha^{m+n-1}, \dots, \alpha, 1)^T$ is in the kernel of the Sylvester matrix: each row times it is $\alpha^k f(\alpha)$ or $\alpha^k g(\alpha)$.

Hint 2. So, as a polynomial in the roots α_i, β_j (the coefficients are elementary symmetric functions of them), the determinant is divisible by each $\alpha_i - \beta_j$, hence by $\prod(\alpha_i - \beta_j)$. Compare degrees and evaluate one coefficient (e.g. $g = x^m$, i.e. all $\beta_j = 0$) to get equality.

Hint 3. For integer polynomials the Sylvester determinant is an integer computed from the coefficients; with leading coefficients a, b it equals $a^m b^n \prod(\alpha_i - \beta_j)$, which is 0 if there is a common complex root.

Problem 10

Hint 1. A transposition of two variables changes the sign of $\Delta = \prod_{i<j}(x_i - x_j)$, so Δ^2 is symmetric; express it through e_1, e_2, e_3 by Theorem 1 (or via $\Delta = \det(x_i^{j-1})$ and $\Delta^2 = \det(V^T V) = \det(p_{i+j})$).

Hint 2. For x^3+ax^2+bx+c: $D = a^2b^2-4b^3-4a^3c-27c^2+18abc$; for x^3+px+q: $D = -4p^3-27q^2$.

Hint 3. $\sigma(\Delta) = \operatorname{sgn}(\sigma)\Delta$ and $\Delta \neq 0$ for distinct roots, so $\Delta = \pm\sqrt{D}$ is fixed exactly by the even permutations. By the Galois correspondence $\Delta \in \mathbb{Q}$ iff Δ is fixed by every element of the Galois group G, i.e. iff $G \subseteq A_n$.